swAV

fold1:
Train samples: 8012
Test samples: 2003
Fold 1 Train Distribution:
{'akiec': 261, 'bcc': 411, 'bkl': 880, 'df': 92, 'mel': 890, 'nv': 5364, 'vasc': 114}
Fold 1 Test Distribution:
{'akiec': 66, 'bcc': 103, 'bkl': 219, 'df': 23, 'mel': 223, 'nv': 1341, 'vasc': 28}

Accuracy Improved (0.8032950574138792 to 0.8302546180728907), Saving the model.....

Evaluation report on Train data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.70 | 0.59 | 0.64 | 261 |
| bcc | 0.75 | 0.74 | 0.74 | 411 |
| bkl | 0.72 | 0.72 | 0.72 | 880 |
| df | 0.70 | 0.21 | 0.32 | 92 |
| mel | 0.68 | 0.51 | 0.58 | 890 |
| nv | 0.90 | 0.96 | 0.93 | 5364 |
| vasc | 0.82 | 0.74 | 0.78 | 114 |
| accuracy | | | 0.85 | 8012 |
| macro avg | 0.75 | 0.64 | 0.67 | 8012 |
| weighted avg | 0.84 | 0.85 | 0.84 | 8012 |

Evaluation report on Test data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.53 | 0.38 | 0.44 | 66 |
| bcc | 0.53 | 0.93 | 0.68 | 103 |
| bkl | 0.85 | 0.53 | 0.65 | 219 |
| df | 0.50 | 0.17 | 0.26 | 23 |
| mel | 0.67 | 0.62 | 0.64 | 223 |
| nv | 0.90 | 0.94 | 0.92 | 1341 |
| vasc | 0.79 | 0.96 | 0.87 | 28 |
| accuracy | | | 0.83 | 2003 |
| macro avg | 0.68 | 0.65 | 0.64 | 2003 |
| weighted avg | 0.83 | 0.83 | 0.82 | 2003 |

Epoch: 5 | train_loss: 0.4220 | train_acc: 0.8461 | test_loss: 0.5073 | test_acc: 0.8303 |

Fold2:
Train samples: 8012
Test samples: 2003
Fold 2 Train Distribution:
{'akiec': 261, 'bcc': 412, 'bkl': 879, 'df': 92, 'mel': 890, 'nv': 5364, 'vasc': 114}
Fold 2 Test Distribution:
{'akiec': 66, 'bcc': 102, 'bkl': 220, 'df': 23, 'mel': 223, 'nv': 1341, 'vasc': 28}
Accuracy Improved (0.81827259111333 to 0.8327508736894658), Saving the model.....

Evaluation report on Train data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.69 | 0.58 | 0.63 | 261 |
| bcc | 0.77 | 0.79 | 0.78 | 412 |
| bkl | 0.77 | 0.77 | 0.77 | 879 |
| df | 0.53 | 0.17 | 0.26 | 92 |
| mel | 0.71 | 0.56 | 0.63 | 890 |
| nv | 0.91 | 0.96 | 0.93 | 5364 |
| vasc | 0.85 | 0.78 | 0.81 | 114 |
| accuracy | | | 0.86 | 8012 |
| macro avg | 0.75 | 0.66 | 0.69 | 8012 |
| weighted avg | 0.85 | 0.86 | 0.85 | 8012 |

Evaluation report on Test data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.48 | 0.61 | 0.54 | 66 |
| bcc | 0.82 | 0.57 | 0.67 | 102 |
| bkl | 0.70 | 0.72 | 0.71 | 220 |
| df | 0.57 | 0.17 | 0.27 | 23 |
| mel | 0.74 | 0.38 | 0.50 | 223 |
| nv | 0.89 | 0.97 | 0.92 | 1341 |
| vasc | 0.69 | 0.89 | 0.78 | 28 |
| accuracy | | | 0.83 | 2003 |
| macro avg | 0.70 | 0.62 | 0.63 | 2003 |
| weighted avg | 0.83 | 0.83 | 0.82 | 2003 |

Epoch: 6 | train_loss: 0.3907 | train_acc: 0.8591 | test_loss: 0.4786 | test_acc: 0.8328 |

Fold 3 sizes:
Train samples: 8012
Test samples: 2003
Fold 3 Train Distribution:
{'akiec': 262, 'bcc': 411, 'bkl': 879, 'df': 92, 'mel': 890, 'nv': 5364, 'vasc': 114}
Fold 3 Test Distribution:
{'akiec': 65, 'bcc': 103, 'bkl': 220, 'df': 23, 'mel': 223, 'nv': 1341, 'vasc': 28}
Accuracy Improved (0.8127808287568647 to 0.8497254118821768), Saving the model.....

Evaluation report on Train data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.87 | 0.89 | 0.88 | 262 |
| bcc | 0.94 | 0.93 | 0.93 | 411 |
| bkl | 0.91 | 0.93 | 0.92 | 879 |
| df | 0.86 | 0.73 | 0.79 | 92 |
| mel | 0.87 | 0.81 | 0.84 | 890 |
| nv | 0.97 | 0.98 | 0.97 | 5364 |
| vasc | 0.93 | 0.91 | 0.92 | 114 |
| accuracy | | | 0.94 | 8012 |
| macro avg | 0.91 | 0.88 | 0.89 | 8012 |
| weighted avg | 0.94 | 0.94 | 0.94 | 8012 |

Evaluation report on Test data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.52 | 0.74 | 0.61 | 65 |
| bcc | 0.74 | 0.79 | 0.76 | 103 |
| bkl | 0.73 | 0.75 | 0.74 | 220 |
| df | 0.32 | 0.39 | 0.35 | 23 |
| mel | 0.79 | 0.45 | 0.57 | 223 |
| nv | 0.92 | 0.95 | 0.93 | 1341 |
| vasc | 0.89 | 0.86 | 0.87 | 28 |
| accuracy | | | 0.85 | 2003 |
| macro avg | 0.70 | 0.70 | 0.69 | 2003 |
| weighted avg | 0.85 | 0.85 | 0.84 | 2003 |

Epoch: 11 | train_loss: 0.1656 | train_acc: 0.9441 | test_loss: 0.4683 | test_acc: 0.8497 |

Fold 4 sizes:
Train samples: 8012
Test samples: 2003
Fold 4 Train Distribution:
{'akiec': 262, 'bcc': 411, 'bkl': 879, 'df': 92, 'mel': 891, 'nv': 5364, 'vasc': 113}
Fold 4 Test Distribution:
{'akiec': 65, 'bcc': 103, 'bkl': 220, 'df': 23, 'mel': 222, 'nv': 1341, 'vasc': 29}
Accuracy Improved (0.8222666000998502 to 0.8322516225661508), Saving the model.....

Evaluation report on Train data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.89 | 0.88 | 0.89 | 262 |
| bcc | 0.94 | 0.93 | 0.93 | 411 |
| bkl | 0.90 | 0.90 | 0.90 | 879 |
| df | 0.82 | 0.67 | 0.74 | 92 |
| mel | 0.87 | 0.81 | 0.84 | 891 |
| nv | 0.96 | 0.98 | 0.97 | 5364 |
| vasc | 0.93 | 0.90 | 0.91 | 113 |
| accuracy | | | 0.94 | 8012 |
| macro avg | 0.90 | 0.87 | 0.88 | 8012 |
| weighted avg | 0.94 | 0.94 | 0.94 | 8012 |

Evaluation report on Test data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.83 | 0.58 | 0.68 | 65 |
| bcc | 0.55 | 0.93 | 0.69 | 103 |
| bkl | 0.84 | 0.52 | 0.64 | 220 |
| df | 0.41 | 0.61 | 0.49 | 23 |
| mel | 0.57 | 0.68 | 0.62 | 222 |
| nv | 0.93 | 0.92 | 0.93 | 1341 |
| vasc | 0.79 | 0.79 | 0.79 | 29 |
| accuracy | | | 0.83 | 2003 |
| macro avg | 0.70 | 0.72 | 0.69 | 2003 |
| weighted avg | 0.85 | 0.83 | 0.83 | 2003 |

Epoch: 12 | train_loss: 0.1683 | train_acc: 0.9412 | test_loss: 0.7117 | test_acc: 0.8323 |

Fold 5 sizes:
Train samples: 8012
Test samples: 2003
Fold 5 Train Distribution:
{'akiec': 262, 'bcc': 411, 'bkl': 879, 'df': 92, 'mel': 891, 'nv': 5364, 'vasc': 113}
Fold 5 Test Distribution:
{'akiec': 65, 'bcc': 103, 'bkl': 220, 'df': 23, 'mel': 222, 'nv': 1341, 'vasc': 29}

Accuracy Improved (0.8167748377433849 to 0.854218671992012), Saving the model.....

Evaluation report on Train data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.82 | 0.82 | 0.82 | 262 |
| bcc | 0.92 | 0.91 | 0.92 | 411 |
| bkl | 0.89 | 0.89 | 0.89 | 879 |
| df | 0.76 | 0.54 | 0.63 | 92 |
| mel | 0.81 | 0.73 | 0.76 | 891 |
| nv | 0.95 | 0.97 | 0.96 | 5364 |
| vasc | 0.93 | 0.95 | 0.94 | 113 |
| accuracy | | | 0.92 | 8012 |
| macro avg | 0.87 | 0.83 | 0.85 | 8012 |
| weighted avg | 0.92 | 0.92 | 0.92 | 8012 |

Evaluation report on Test data:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| akiec | 0.66 | 0.69 | 0.68 | 65 |
| bcc | 0.67 | 0.79 | 0.72 | 103 |
| bkl | 0.75 | 0.75 | 0.75 | 220 |
| df | 0.59 | 0.43 | 0.50 | 23 |
| mel | 0.77 | 0.49 | 0.60 | 222 |
| nv | 0.91 | 0.95 | 0.93 | 1341 |
| vasc | 0.85 | 0.79 | 0.82 | 29 |
| accuracy | | | 0.85 | 2003 |
| macro avg | 0.74 | 0.70 | 0.71 | 2003 |
| weighted avg | 0.85 | 0.85 | 0.85 | 2003 |

Epoch: 9 | train_loss: 0.2216 | train_acc: 0.9196 | test_loss: 0.4307 | test_acc: 0.8542 |